

National University of Computer and Emerging Sciences

Q4 [10+5 marks]

Consider a Recommendation system where there exist two ranking systems. One is ranked by top movie critics and the other is user based. Write a **C program** to implement a simple movie recommendation system using file handling. The program should perform the following tasks:

Requirements:

1. Book Data Storage:

- Create a file (movie.bin) that contains a list of movies. Each line in the file should represent a movie in the format:
<MovieID>,<Title>,<Director>,<Genre>,<Rating_IMDB>,<Rating_Rotten_Tomatoes>
- Example: 101,The_Terminal, Comedy, 7.4, 61% (Create at least 5 listings)

2. Search and Recommend:

Allow the user to search for Movies based on:

IMDB Rating: On a scale of 1-10

Rotten Tomatoes Rating: On a Scale of 1-100.

3. Add a New Movie:

Provide an option for the user to add a new movie to the file. Ensure the new entry maintains the same format.

4. Display All Movies:

Display all movies stored in the file in a tabular format.

Note: Store the file in the .bin format and use fseek to traverse into the file.

Some Dummy Data to help

101,The_Terminal, Comedy, 7.4, 61%
102,The Polar Express, Kids & Family, 6.6, 56%
103,The Green Mile, Drama/Fantasy, 8.6, 79%
104,Toy Story, Animation, 8.3, 100%
105,Sully, Drama/Biography, 7.4, 85%

(I could have selected better movies so write it down on the line below the best movie you ever watched).

Programming Fundamentals Lab (CL1002)

Date: 05/12/2024

Course Instructor(s)

Ms Bushra

Lab Final Exam

Total Time: 120 minutes

Total Marks: 50

Total Questions: 04

Semester: FL-2024

Campus: Karachi

Dept: Computer Science

Name: _____

Roll No: _____

CLO # 1: Demonstrate the ability to define and implement modular code using functions, and apply recursion to solve problems effectively, emphasizing problem decomposition and logical thinking.

Q1. [10 marks]

Write a recursive program in C that prints a sideways triangle pattern of stars (*) given a maximum width. The triangle should increase in width from 1 star up to the maximum width and then decrease back down to 1 star.

Requirements:

1. Use recursion to solve the problem.
2. Create a helper recursive function to print stars in a single row.
3. Do not use loops (for, while) in the solution.

Hint: You would require two functions to solve this.

Output:

```
*
**
***
****
***
**
*
```

CLO # 2: Utilize pointers to manipulate memory addresses and dynamically allocate, manage, and deallocate memory to create efficient and flexible programs.

Q2. [6+4 marks]

Write a C program to dynamically manage a collection of strings entered by the user. The program should use pointers and dynamic memory allocation to achieve the following:

Requirements:

1. Allow the user to input strings of varying lengths. Each string should be dynamically allocated using malloc.
2. Store the strings in a dynamically allocated array of char* pointers, starting with an initial capacity of 3 strings.
3. If the array becomes full, double its capacity using realloc to accommodate more strings.

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4. Provide the following functionality:

- Add a new string to the collection.
- Display all stored strings.
- Delete a specific string from the collection by its index.
- Free all allocated memory when the program terminates.

Output:

Enter a string (or "STOP" to finish): Hello
Enter a string (or "STOP" to finish): World
Enter a string (or "STOP" to finish): Dynamic
Enter a string (or "STOP" to finish): Memory
Array resized to capacity: 6
Enter a string (or "STOP" to finish): STOP

CLO # 3: CLO # 3: Apply 2D arrays to represent and manipulate structured data and handling techniques for persistent data storage and retrieval in programs.

Q3 [10+5 marks]

In Q1 you already made a sideways triangle using recursion. You are now tasked with making a sideways triangle in a 2D array. Note that the triangle size would be user defined so you can store it in an array. Once fully stored Flip the triangle Horizontally within the array. You can utilize another array to store some elements.

Output:

(In array)

```
*  
**  
***  
****  
****  
***  
**  
*
```

(In array)

```
*  
 **  
  ***  
   ****  
  ****  
 ***  
**  
*
```

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